I.

An Old and a New Tradition

WHEN one speaks today of modern physics, the first thought is of atomic weapons. Everybody realizes the enormous influence of these weapons on the political structure of our present world and is willing to admit that the influence of physics on the general situation is greater than it ever has been before. But is the political aspect of modern physics really the most important one? When the world has adjusted itself in its political structure to the new technical possibilities, what then will remain of the influence of modern physics?

To answer these questions, one has to remember that every tool carries with it the spirit by which it has been created. Since every nation and every political group has to be interested in the new weapons in some way irrespective of the location and of the cultural tradition of this group, the spirit of modern physics will penetrate into the minds of many people and will connect itself in different ways with the older traditions. What will be the outcome of this impact of a special branch of modern science on different powerful old traditions? In those parts of the world in which modern science has been developed the primary interest has been directed for a long time toward practical activity, industry and engineering combined with a rational analysis of the outer and inner conditions for such activity. Such people will find it rather easy to cope with the new ideas since they have had time for a slow and gradual adjustment to the modern scientific methods of thinking. In other parts of the world these ideas would be confronted with the religious and philosophical foundations of the native culture. Since it is true that the results of modern physics do touch such fundamental concepts as reality, space and time, the confrontation may lead to entirely new developments which cannot yet be foreseen. One characteristic feature of this meeting between modern science and the older methods of thinking will be its complete internationality. In this exchange of thoughts the one side, the old tradition, will be different in the different parts of the world, but the other side will be the same everywhere and therefore the results of this exchange will be spread over all areas in which the discussions take place.

For such reasons it may not be an unimportant task to try to discuss these ideas of modern physics in a not too technical language, to study their philosophical consequences, and to compare them with some of the older traditions.

The best way to enter into the problems of modern physics may be by a historical description of the development of quantum theory. It is true that quantum theory is only a small sector of atomic physics and atomic physics again is only a very small sector of modern science. Still it is in quantum theory that the most fundamental changes with respect to the concept of reality have taken place, and in quantum theory in its final form the new ideas of atomic physics are concentrated and crystallized. The enormous and extremely complicated experimental equipment needed for research in nuclear physics shows another very impressive aspect of this part of modern science. But with regard to the experimental technique nuclear physics represents the extreme extension of a method of research which has determined the growth of modern science ever since Huyghens or Volta or Faraday. In a similar sense the discouraging mathematical complication of some parts of quantum theory may be said to represent the extreme consequence of the methods of Newton or Gauss or Maxwell. But the change in the concept of reality manifesting itself in quantum theory is not simply a continuation of the past; it seems to be a real break in the structure of modern science. Therefore, the first of the following chapters will be devoted to the study of the historical development of quantum theory.